

# Brief Report: Channel Estimation with LS, LMMSE and CNN

## 1. Objective

Implement and compare a basic channel estimation process (LS and LMMSE) with a simple CNN-based estimator, using NMSE and BER as the main metrics.

## 2. System Setup

- **Link:** OFDM with 64 subcarriers and an all-pilot block (QPSK pilots).
- **Channel:** Rayleigh multipath with 8 taps and an exponential power-delay profile (decay 0.5), assumed shorter than the cyclic prefix.
- **Noise:** AWGN, evaluated at SNR 0 to 30 dB in 5 dB steps.

## 3. Methods

- **LS:**  $H = Y/X$ . It needs no prior knowledge and does no denoising.
- **LMMSE:**  $H = R(R + \sigma^2 I)^{-1} H_{LS}$ . It uses the exact channel covariance and the SNR.
- **CNN:**
  - A 5-layer 1D CNN with 64 filters, kernel size 5 and circular padding.
  - Input is Re/Im of the LS estimate plus an SNR channel; it predicts a residual correction on LS.
  - Trained for 30 epochs with Adam and MSE loss on freshly generated data at random SNRs from 0 to 30 dB.

## 4. Metrics

- **NMSE:**  $\|H - \hat{H}\|^2 / \|H\|^2$ , in dB.
- **BER:** QPSK data equalized with ZF using each estimate, with perfect CSI as a reference.
- **Also reported:** parameter count and inference time.

## 5. Results

| SNR   | NMSE LS          | NMSE LMMSE       | NMSE CNN            |
|-------|------------------|------------------|---------------------|
| 0 dB  | $\approx 0$ dB   | $\approx -10$ dB | $\approx -9.66$ dB  |
| 10 dB | $\approx -10$ dB | $\approx -19$ dB | $\approx -19.5$ dB  |
| 30 dB | $\approx -30$ dB | $\approx -39$ dB | $\approx -35.62$ dB |

NMSE values for the CNN are read from the plot.

- LS NMSE equals  $-\text{SNR}$ , as expected.
- LMMSE gains about 9 dB over LS, matching the  $L/N = 8/64$  theory.
- The CNN comes close to LMMSE at low and mid SNR, and the gap grows to about 3 to 4 dB at high SNR.
- In BER, LMMSE, CNN and perfect CSI nearly overlap and all beat LS.

## 6. Observations

- The CNN reaches near-LMMSE performance without being given the covariance. It learned the channel structure from data.
- The CNN lags at high SNR because plain MSE loss is dominated by low-SNR training samples.
- The training loss is above the validation loss because training spans 0 to 30 dB while validation is at 10 dB only. This is not overfitting.
- LMMSE has an advantage here because it is given the exact covariance, which is rarely known in practice.

## 7. Limitations and Next Steps

- The estimators have been tested only on the same channel statistics they were designed or trained for.
- **Robustness test:** train at DECAY = 0.5 and test with different delay profiles or tap counts. LMMSE keeps its fixed covariance, so this shows where it breaks and how the CNN generalizes.
- **Loss function:** use a normalized per-sample loss to improve high-SNR accuracy.
- **Pilot pattern:** try comb-type pilots with interpolation as the baseline.

*Code: MLProject.ipynb attached.*